Source EOS versus Cowling solver adapter

The upstream EOS generator's master-function reconstruction already produces first-law-consistent thermodynamic knots and smooth crust-core matching bridges. The adapter addresses a different requirement: the Cowling solver uses a native cubic-Hermite P(mu_B) interpolator with exact derivative n_B=P' and requires positive, causal thermodynamics throughout each cell.

For the 234 selected EOSs, source q_i=n_i/mu_i is already nondecreasing. The weighted PAV stage pooled zero knots and modified q only at roundoff. The actual representation change was to take q linear between knots, integrate P'=n=mu q, and derive epsilon=mu n-P. This yields an exact cubic P(mu) per cell and c_s^2=q/(q+mu q').

Saved cohort diagnostics: - max relative density change: 3.1769e-16; - max abs pressure change / P_max: 1.2931e-3; median 5.0781e-4; - max abs energy change / (mu n): 8.3590e-4; median 3.6598e-4; - max adapter first-law residual: 2.163e-16; - projected native cell audit: PASS for all 234.

These changes are small but not zero. They are a difference between EOS representations and belong in any systematic error budget. The original joined EOS tables are retained unchanged. The q-linear adapter is not to be reported as a new microscopic EOS or as proof that all alternative interpolants are causal.